

What Drives RF TEST EQUIPMENT



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From defence radars, magnetic resonance imaging (MRI) and mobile communications, to microwave ovens and Wi-Fi, radio frequency (RF) signals are present all around you. While this is a testament to the importance of RF, it also shows why it is critical to make sure that the numerous RF applications in the world work perfectly as designed, and are also able to coexist with each other.

This article takes a look at some of the latest tools available to solve RF design challenges.

Get a comprehensive view of the device under test

Field-programmable gate array based signal processing continues to benefit test equipment in terms of speed and capacity. This has allowed RF instrument manufacturers to integrate more functionality into a single box.

Dr Jan Benhelm, head of marketing, Zurich Instruments AG, says, "Combining this digital-signal-processing capability with the finest analogue front-ends allows us to design instruments with unprecedented performance, lowest input noise and highest dynamic range, and at the same time, offering a complete toolset for time and frequency domain analysis has become possible."

"Zurich Instruments UHFLLI lock-in amplifier with a bandwidth of 600MHz leverages this to offer the fastest lock-in amplifier on the market today, but it also comprises an oscilloscope, digitiser, boxcar average, real-time arithmetic unit, parametric sweeper, FFT spectrum analyser, 4x PID controllers, 4x auxiliary outputs and 2x signal generators."

With these multiple functions at hand, the user can simultaneously analyse signal inputs in different ways and get a comprehensive view of the setup or device under test. One of the biggest advantages of digital signal processing over analogue is that you can

copy a signal without loss of signal-to-noise ratio. This opens a multitude of possibilities limited only by the amount of available memory and signal-processing capacity.

Software to help you set up

Instrument-control tools like LabOne work across all instrument platforms to provide a high level of support to the user when setting up demanding measurements. One prominent example is the feedback-loop operation. "It does not matter if you are testing your new micro-electro-mechanical systems (MEMS) gyro or phase-locking two lasers, LabOne will help you to find the right proportional-integral-derivative (PID) parameters and characterise the feedback-loop performance. If needed, autotune feature will automatically try to reduce the standard deviation of the error signal by adjusting PID parameters," adds Dr Benhelm.

Model functions of the transfer function and step response are provided that can easily be compared to measurements with the parametric sweeper or the built-in oscilloscope. Some application fields are MEMS characterisation, fluorescence lifetime spectroscopy, THz spectroscopy, synchrotron experiments, mass spectroscopy and pulsed radar applications.

Newer equipment improve productivity

The new source-phase-control option is introduced on high-performance network analysers, which aim to help study device response under varying phases. This option would prove to be useful for engineers carrying out active load pull measurements on amplifiers.

Another recent development is the ability to control the phase of multiple signal sources (internal and external) for testing differential IQ modulators. This used to be a lot more difficult earlier.

Ashutosh Dwivedi, application engineer, Keysight Technologies, says, "High-end network analysers are targeted at customers that include those working on wafer-level device characterisation and device mod-

Rohde & Schwarz
vector network analyser

